Media-in-the-loop holography: designing photopolymer exposures through a differentiable recording model

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Abstract: A write-once volume photopolymer must be exposed before the medium distorts the target, and the error is measurable only once it is permanent. We optimize the exposure through a differentiable recording model: non-local polymerization-driven diffusion read out by beam propagation. Over 30 spatial-frequency and contrast-budget cells the designed exposure beats a media-blind optimizer in 29, by up to 2.52 dB, and removes every exposure falling more than 2 dB short of an oracle reference. The gain needs a nonlinear dose response, depends most on the index budget, and vanishes once the fringes tilt. We give the calibration measurements required and when not to use the method.

1. Introduction

A laboratory recording a volume hologram in a photopolymer faces a question that computer-generated holography on a spatial light modulator does not. The medium is written once. Whatever the exposure delivers is what the material keeps, and the reconstruction error can be measured only after it is permanent. Three practical questions follow. When is it worth designing the exposure through a model of the recording, rather than treating the medium as a linear transfer? When does that design stop helping? And what must be measured to calibrate the model before trusting it? This paper answers all three for one medium family, in simulation, and reports the boundary of each answer.

On a rewritable modulator the corresponding problem is solved by measurement. Camera-in-the-loop (CITL) optimization records the reconstruction and back-propagates the residual into the display model [1]; related approaches refine the model with Wirtinger gradients [2], learned proxies [3] or time-multiplexed control [4]. Every one of them needs to rewrite the modulator and try again.

Volume photopolymers such as Bayfol HX [5], acrylamide/PVA and PQ/PMMA [6] record a permanent index distribution through photopolymerization and mass transport, and the recorded index is not a scaled copy of the exposure: monomer diffusion [7,8], non-local chain growth [9,10], dye bleaching and a saturating index response act nonlinearly [11]. A CITL-style measurement therefore arrives too late, and the recording model has to carry the correction with no feedback at all.

Differentiable models of a fabrication process already sit inside optical design loops: neural lithography learns a photolithography simulator for diffractive optics [12], and adjoint inverse design treats a physical forward model as a differentiable layer [13]. Concurrently, Wechsler et al. [14] coupled a differentiable wave-optical model to a photochemical model for holographic volumetric printing and demonstrated printed parts, an experiment this study does not attempt; we make no priority claim for differentiable recording models in a design loop.

What this paper adds is a characterization, with the numbers a laboratory would need before committing material. We measure how much an exposure optimized through a differentiable recording model ("media-in-the-loop", MIL) improves on the same optimizer run with a linear, media-blind recording assumption, and then map where that improvement lives: across spatial

frequency and contrast budget (Section 4.1), against a reference that says how much of the achievable quality is being thrown away (Section 4.2), across the recording mechanisms that could be responsible (Section 4.5), and against fringe slant, where it stops (Section 4.6). Section 4.8 ranks the model parameters by how much the design gain depends on getting each one right, and Section 6 turns that ranking into a calibration protocol: which measurements are needed, in what order, and when to leave the method alone.

The recording model (non-local polymerization-driven diffusion, NPDD), the readout (split-step beam propagation) and the optimizer are all established; the contribution is their combination into an exposure-design loop and its characterization. Every headline number is a paired difference (MIL minus media-blind design) with both arms designed and scored on the same simulated medium, the "twin". An exposure-independent calibration error of the twin cancels in that difference; an exposure-dependent one does not, so we also design on the nominal twin and record on miscalibrated ones (Sections 4.7 and 4.8) and compare the twin with published data (Section 5). All results are simulation outputs of this study unless cited, and every number is generated by script from released per-job result files. This is the second part of a two-part study; Part 1 characterized phase-only CGH degradation with idealized modulators [15].

2. Model

Table 1 lists every constant; the released code is the reference implementation.

2.1. Exposure and feasible set

The design variable is the delivered exposure $E(x) \geq 0$ on the transverse coordinate x, constrained to a mean dose E_0 and a peak-to-mean contrast budget B_c:

$$\mathcal{E}(E_0, B_c) = \left\{ E : \; E \geq 0, \; \overline{E} = E_0, \; \max_x E(x)/\overline{E} \leq B_c \right\}. \tag{1}$$

B_c is the quantity a recording setup actually controls, through the fringe visibility and the beam ratio. After every optimizer step the exposure is projected onto $\mathcal{E}$ by 20 alternating passes of dose rescaling, $E \leftarrow E\,E_0/(\overline{E} + \epsilon)$, and peak clipping, $E \leftarrow \min(E, B_c E_0)$. Both operations are elementwise and differentiable; the residual relative constraint violation after 20 passes is below 10^{-6} on the tested targets.

2.2. Recording: non-local polymerization-driven diffusion

Free monomer $u(x,t)$ ($u(x,0) = 1$), polymer $N(x,t)$ ($N(x,0) = 0$) and unbleached dye $d(x,t)$ ($d(x,0) = 1$) evolve as in the NPDD model [9], itself built on the local model of Zhao and Mouroulis [7]:

$$\partial_t u = \partial_x\big(D_{\text{eff}}(N)\,\partial_x u\big) - G_\sigma * (Fu), \tag{2}$$

$$\partial_t N = G_\sigma * (Fu), \tag{3}$$

$$\partial_t d = -k_{\text{bleach}}\, I\, d, \tag{4}$$

with $F = \kappa\,(Id)^\gamma$, recording intensity $I \propto E$, a Gaussian non-local kernel G_σ of width σ ($\widehat{G_\sigma}(K) = e^{-K^2\sigma^2/2}$) and $D_{\text{eff}}(N) = D_0 e^{-\alpha_D N}$. The kernel acts on the production term Fu, so that polymer initiated at x' is deposited at x [9]. The recorded index modulation is

$$\Delta n(x) = \Delta n_{\max} \tanh\big(\beta N(x)\big), \qquad \beta = 1.5. \tag{5}$$

The saturating map and the value of β are modelling choices of this work, held fixed in every fit. Three mechanisms saturate the dose response: monomer depletion ($u \to 0$), dye bleaching ($d \to 0$) and the tanh map; Section 4.5 separates them.

2.3. Small-signal transfer function and a first design rule

Before any optimization, the recording equations already say roughly where designing through them can help. Take $I(x) = I_{\text{mean}}(1 + m\cos Kx)$ with $m \ll 1$, neglect dye depletion ($d \approx 1$) and hold $D_{\text{eff}} = D_0$, so that Eqs. (2)–(4) are linearized about the start of recording. The uniform fields obey $\dot{u}_0 = -F_0 u_0$ with $F_0 = \kappa I_{\text{mean}}^{\gamma}$, and eliminating the first-order Fourier component quasi-statically gives

$$H(K) = \frac{\exp(-K^2\sigma^2/2)}{1 + D_0 K^2/F_0}. \tag{6}$$

Non-locality rolls the recorded modulation off as a Gaussian in $K\sigma$ and diffusion as a Lorentzian in $K\sqrt{D_0/F_0}$. Undoing that roll-off asks for a pre-boost $1/H(K)$ of the target's spatial-frequency content, and the contrast budget caps how much pre-boost an exposure can carry: beyond some $K_c(B_c)$ the required boost exceeds the realizable contrast, and the heuristic predicts the advantage should change sign there.

Equation (6) is a motivating argument, not a validated transfer function. It does not match the full simulator, while an independent numerical linearization of the same equations does (Supplement 1, Section S1). It is used here only to place the K_c markers in Fig. 1(a) and to define the linear pre-compensation baseline of Section 3; no result depends on its being exact, and Section 4.1 reports where the data depart from it.

2.4. Integration and gradients

Equations (2)–(4) are integrated over $t_{\text{total}} = 10\,\text{s}$ in $n_{\text{steps}} = 300$ first-order implicit–explicit steps. The explicit reaction step evaluates the non-local production by fast Fourier transform, caps consumed monomer at the monomer available (so $u \geq 0$ and $u + N$ is conserved to rounding error) and updates d exactly for fixed I. The implicit diffusion step applies $\exp(-\bar{D}K^2\,\mathrm{d}t)$ in Fourier space with $\bar{D}$ the spatial mean of D_{eff}; this approximation to Eq. (2) has a one-step error of about 0.1% of the scale of u at a representative late state, and its accumulation over a trajectory was not bounded. Gradients are the discrete adjoint of this integrator, obtained by reverse-mode automatic differentiation [16, 17], and agree with central finite differences to a maximum relative error of 6.5×10^{-6} over 6 random directions through the full chain (Supplement 1, Section S10).

2.5. Readout: split-step beam propagation

The recorded profile $\Delta n(x)$ is extruded uniformly through a slab of thickness T and read out at wavelength λ by scalar split-step beam propagation [18]: $n_z = 128$ slices, each applying the phase $\exp[ik_0\Delta n\,\mathrm{d}z(1-s)]$ ($k_0 = 2\pi/\lambda$, shrinkage s) followed by band-limited angular-spectrum propagation [19], then one angular-spectrum step to the reconstruction plane, where the intensity is the reconstruction R. The periodic window represents an infinite grating. Slanted fringes ($x - z\tan\phi = \text{const}$) are represented by a per-slice lateral Fourier shift. All headline results use an unslanted grating ($\phi = 0$), the geometry in which scalar theory is closest to rigorous coupled-wave analysis (RCWA) [20] (Section 4.6). Between $n_z = 128$ and 256 the paired gain changes from 0.915 to 0.894 dB at $K = 1.96\,\text{rad/µm}$ and from 0.900 to 0.928 dB at $K = 3.93\,\text{rad/µm}$ (one seed, Supplement 1, Section S3). The Klein–Cook parameter $Q = 2\pi\lambda T/(n_0\Lambda^2)$, $\Lambda = 2\pi/K$ [21], crosses 10 at $K \approx 2.79\,\text{rad/µm}$, so the two lowest tested K lie in the intermediate regime. Kogelnik theory [22] is used only as a closed-form cross-check.

2.6. Loss and optimizer

With target intensity I_t on n_x samples, the loss is the scale-invariant mean-squared error

$$\mathcal{L}(R, I_t) = \frac{1}{n_x}\left\|\alpha^\star R - I_t\right\|_2^2 \Big/ \max(I_t)^2, \qquad \alpha^\star = \langle R, I_t\rangle/\langle R, R\rangle, \tag{7}$$

and we report $\mathrm{PSNR}_{\mathrm{si}} = -10\log_{10}\mathcal{L}$, a monotone transform of the optimized objective. The optimizer is Adam [23] (learning rate 5×10^{-2}, 800 iterations, convergence tolerance 10^{-4}) on the parameterization $E = \mathrm{softplus}(\theta) + 10^{-6}$. The random seed enters only through the initialization $\theta_0 = 10^{-2}z$, $z \sim \mathcal{N}(0, I)$, so seed-to-seed spread measures sensitivity to the starting point, not model or parameter uncertainty.

Table 1. Default medium and simulation parameters. Medium values are illustrative acrylamide/PVA-like choices inside cited ranges, not a refit of one published medium.

Parameter	Default	Source
D_0	$0.1\,\mu\mathrm{m}^2/\mathrm{s}$	illustrative; AA/PVA literature value 0.01 [8]
σ	$0.08\,\mu\mathrm{m}$	illustrative; literature range 0.01–0.04 μm [10]
κ	2.0	calibration choice (cures the medium to $N \approx 1$ at unit dose)
γ	1.0	within the literature range 0.5–1.0 [9]
$\Delta n_{\max}$	3.5×10^{-3}	within the AA/PVA range 1.5–5×10^{-3}
k_{bleach}, α_D	0.2, 1.0	order of magnitude, not independently sourced
β	1.5	shape constant of Eq. (5) (this work)
s	0.5%	placeholder of typical AA/PVA magnitude
T, n_0, λ, ϕ	30 μm, 1.5, 405 nm, 0°	operating point; slant swept in Section 4.6
Grid	$n_x = 1024$, $\mathrm{d}x = 0.05\,\mu\mathrm{m}$	51.2 μm periodic window; $n_z = 128$

3. Methods compared and benchmark

All methods share the exposure parameterization, the projection onto $\mathcal{E}$ and scoring on the full twin. **MIL** optimizes Eq. (7) with the twin (NPDD and beam propagation) inside the loss. **Media-blind SGD (BSGD)** runs the same optimizer under the linear recording assumption $\Delta n = \Delta n_{\max}(E - \overline{E})$ without shrinkage and scores its exposure on the twin; it is the design a laboratory produces when it treats the medium as a linear transfer, and it is the reference every gain in this paper is measured against. **Gerchberg–Saxton (GS)** is alternating-projection phase retrieval [24, 25], its phase converted to an exposure by the same linear assumption. **Linear pre-compensation (LPC)** boosts each spatial-frequency component of the target by $1/H(K)$ from Eq. (6) and then projects, so it is the closed-form use of the Section 2.3 design rule. **Gamma pre-compensation (GPC)** inverts a fitted pointwise saturating map. **TV-regularized SGD (RSGD)** is BSGD with a total-variation penalty whose weight was tuned on the twin per budget; the tuned weight is zero at all three budgets, so RSGD reduces to BSGD. The **saturation-only surrogate (SAT)** optimizes against a pointwise saturating model with all transport removed, calibrated to the twin by one fitted parameter (NRMSE 0.103); it asks whether a laboratory could skip the transport physics. The **constrained oracle (ORC)** optimizes against an ideal linear medium under the same exposure constraints; it is a gradient-optimized reference, not a proven bound.

The benchmark target is a binary bar pattern at spatial frequency K, with periods rounded to an even number of pixels so that each nominal K is rendered exactly. The main grid covers 10 spatial frequencies, $K \in \{1.96, 2.62, 3.49, 3.93, 4.19, 4.83, 5.71, 6.98, 8.98, 15.71\}$ rad/µm, at budgets $B_c \in \{2, 4, 8\}$, i.e. 30 cells, each with every method complete. Ablation and robustness studies use $K = 1.31$, 3.93 and 5.24 rad/µm at $B_c = 2$. Iterative methods run with three seeds and deterministic ones (GS, LPC, GPC) once. The paired gain of a cell is the mean over seeds of $\mathrm{PSNR}_{\mathrm{si}}(\mathrm{MIL}) - \mathrm{PSNR}_{\mathrm{si}}(\mathrm{BSGD})$ at equal seed; intervals are two-sided 95% Student-t intervals over seeds unless a bootstrap [26] is stated. All simulations ran on one laptop (NVIDIA RTX 3050 GPU; the factorial, compute-matched and parameter-sensitivity jobs on its CPU).

4. Results

4.1. *Where designing through the model pays off*

The paired gain is positive in 29 of 30 cells and its 95% interval excludes zero in 27 (Fig. 1(a)). The largest cell means are 1.42, 2.13 and 2.52 dB at $B_c = 2$, 4 and 8, all at $K \leq 2.62$ rad/µm; the grid means are 0.40/0.70/0.88 dB. At the highest tested $K = 15.71$ rad/µm the gain is 0.01, 0.04 and 0.05 dB, which is to say nothing. One cell is negative: $K = 4.19$ rad/µm at $B_c = 2$, -0.03 dB (95% interval [-0.05, -0.005]). The seed intervals are narrow because the seed changes only the initialization (Sections 4.7 and 4.8 treat model error). Diffraction efficiency shows the same budget ordering (mean paired gain 1.80/3.00/3.94 percentage points) but 15 of 90 per-seed pairs are negative (minimum -0.251 points).

The heuristic of Section 2.3 predicts a sign change at $K_c = 4.22$, 6.96 and 9.33 rad/µm (dashed lines in Fig. 1(a)). At $B_c = 2$ the gain crosses zero at $K^* = 4.18$ rad/µm (linear interpolation), close to K_c, but is positive again at every larger tested K, and at $B_c = 4$ and 8 it never crosses. The data therefore show a decay of the gain with K, not monotone at this sampling, rather than the persistent sign change the heuristic predicts. The constrained oracle's own advantage over BSGD falls with K as MIL's does, which we read, without a separate test, as a shrinking ceiling for any exposure-constrained method rather than a failure specific to MIL.

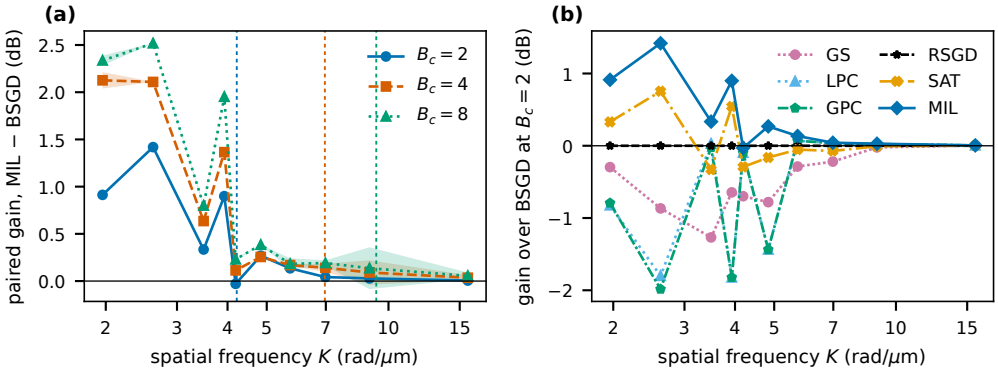


Fig. 1. (a) Paired gain (MIL − BSGD) versus spatial frequency for three contrast budgets; shading, 95% t-intervals over three seeds; dashed lines, heuristic $K_c(B_c)$ from Eq. (6). (b) Gain of every non-oracle method over BSGD at $B_c = 2$ (seed means); RSGD coincides with BSGD.

Figure 2 shows single-seed reconstructions at the three K of the ablation studies; Supplement 1, Fig. S1, shows the corresponding exposures and index profiles. At the lowest K MIL recovers most of the bar contrast that BSGD loses; at the highest, both reconstructions are similarly shallow.

Figure 3(a) maps the same 30 cells as a single picture, and its practical reading is that the benefit is concentrated at the low-frequency, high-budget corner. A larger contrast budget buys more, and a higher spatial frequency takes it away: the advantage is substantial below about 3 rad/µm and is not resolved from zero at the two highest K at $B_c = 4$ and 8.

4.2. *How much material a media-blind design wastes*

Each exposure of a write-once medium consumes material, so average quality is the wrong figure of merit on its own: what matters to a laboratory is how often a recording comes out far worse than it needed to. An exposure fails if its PSNR is more than a threshold below the constrained oracle's in the same cell (Table 2; the 2 dB threshold was fixed before the results were computed).

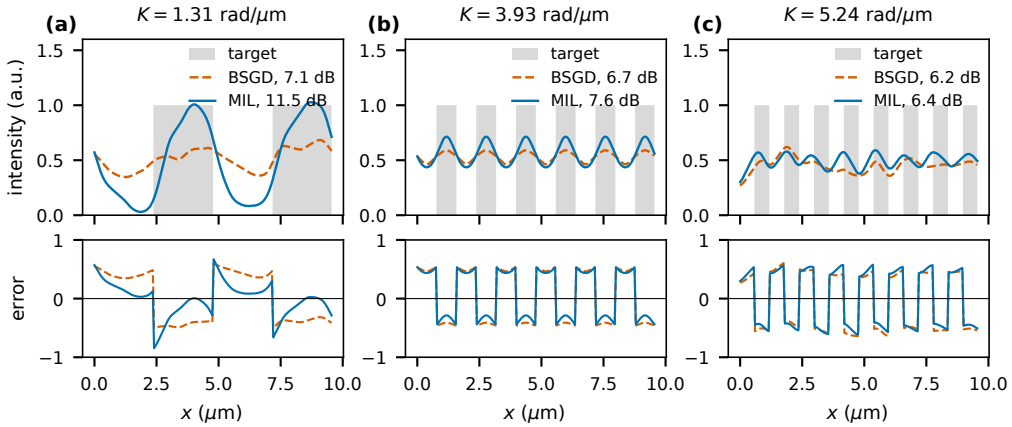


Fig. 2. Target, BSGD and MIL reconstructions (top) and errors (bottom) at $B_c = 2$, seed 0, over a 9.6 μm section of the window; legends give $\mathrm{PSNR}_{\mathrm{si}}$. Single-seed illustrations; Fig. 1 carries the statistics.

At 2 dB, 37% of BSGD cells and 0% of MIL cells fail, and at 3 dB 17% and 0%; at 1 dB the rates are similar (60% and 57%). MIL removes the large shortfalls, not the small ones, which is the shape of benefit that matters when each attempt costs a film. The rates are relative to a numerical reference over 30 cells, not to a proven optimum or over physical trials.

Table 2. Failure rate: percentage of the 30 grid cells in which a method's seed-mean PSNR is more than the column threshold below the constrained oracle's.

Threshold	GS	LPC	GPC	BSGD	RSGD	SAT	MIL
1 dB	60	60	60	60	60	60	57
2 dB	53	47	47	37	37	33	0
3 dB	33	33	30	17	17	0	0

4.3. *Simpler alternatives, and why they are not enough*

At $B_c = 2$ (Fig. 1(b)), GS (-1.27 to 0.00 dB) never improves on BSGD and LPC (-1.82 to 0.07 dB) improves on it by at most 0.07 dB; GPC (-1.98 to 0.07 dB) differs from LPC by at most 0.18 dB, consistent with both hitting the same contrast clip. LPC is the closed-form use of Eq. (6), so its failure is further evidence that the small-signal picture does not describe this regime.

The saturation-only surrogate tests whether a laboratory could model the dose response and ignore transport. Averaged over K, its gain is 18%, 28% and 42% of MIL's at $B_c = 2$, 4 and 8, and by cell it recovers 36–44% of MIL's gain at $K = 1.96$ rad/µm and 53–66% at $K = 2.62$ rad/µm but scores below BSGD in 12 of the 29 cells where the ratio is defined. A pointwise surrogate is therefore usable only at the lowest spatial frequencies; elsewhere the full recording model is needed for any gain at all.

4.4. *Compute-matched control*

MIL's per-iteration cost is about 21.7× that of BSGD (one wall-clock measurement), so a fair question is whether the baseline simply needed longer. With 21.7× more BSGD iterations at the

three ablation K the paired gain is 3.19 [2.94, 3.43], 0.81 [0.81, 0.81] and 0.12 dB [0.12, 0.13], against 4.41, 0.90 and 0.11 dB at equal iteration count: the extra iterations narrow the gap at the two lower K but it stays positive at all three. Wall-clock time is a noisy proxy for compute, so this tests a much larger baseline budget rather than an equal-FLOP comparison.

4.5. Which recording nonlinearity produces the gain

Removing one mechanism at a time (Table 3, top) leaves a positive gain at every K, and which removal matters depends on K: linearizing the index map raises the gain at $K = 1.31$ and 5.24 but lowers it at 3.93 rad/µm. Because three mechanisms saturate the dose response, removing one leaves the others, so we also ran all 2^3 on/off combinations of monomer depletion (removed by holding $u = 1$), dye bleaching ($k_{\text{bleach}} = 0$) and the tanh map, at a matched operating point (Table 3, bottom; all twelve conditions, including the unmatched ones, in Supplement 1, Section S2). Removing monomer depletion at fixed κ raises the polymer cured by unit dose from $N \approx 1$ to $N \approx 8.7$, so these conditions are run with κ rescaled to restore $N(E{=}1) = 1$. A final control removes all three and sets $\kappa = 1/(1.5\, t_{\text{total}})$, which makes the recording BSGD's own linear model up to non-local blur and shrinkage. The factorial ran on CPU and reproduces the GPU full-model values to within 0.001 dB.

Table 3. Paired gain (dB) under mechanism ablations, $B_c = 2$, mean of three seeds. Top: one mechanism removed. Bottom: saturating mechanisms isolated at a matched operating point, $N(E{=}1) = 1$, and the slope-matched linear control; blur, diffusion and shrinkage are kept. Intervals: Supplement 1, Figs. S2–S3.

Condition	$K = 1.31$	$K = 3.93$	$K = 5.24$
Full model	4.41	0.90	0.11
No non-locality ($\sigma = 0$)	4.47	0.82	0.05
No diffusion ($D_0 = 0$)	4.93	1.14	0.10
No dye bleaching ($k_{\text{bleach}} = 0$)	4.36	0.84	0.11
No index saturation (tanh linearized)	5.39	0.61	0.66
Only monomer depletion	5.75	1.73	1.26
Only dye bleaching	1.98	0.42	-0.02
Only tanh map	2.80	0.53	0.12
Dye bleaching and tanh map	3.64	0.72	0.08
Linear, slope-matched	-0.00	0.00	0.10

Three observations follow. First, when the recording is the media-blind model up to blur and shrinkage, MIL gains essentially nothing at the two lower K, so the gain there requires a nonlinear dose response; at $K = 5.24$ rad/µm the linear control's gain (0.10 dB) is comparable to the full model's (0.11 dB), so the small gain there may come from the blur and shrinkage that BSGD ignores. Second, each saturating mechanism alone produces a gain at the two lower K, so no single mechanism is necessary. Third, monomer depletion alone gives a larger gain than the full model at every K, so the mechanisms' effects do not add. For a laboratory this is the encouraging result: the method does not require identifying which nonlinearity a given medium has, only that it has one and that the model captures it. The matching criterion $N(E{=}1) = 1$ is our choice; another normalization could reorder the single mechanisms, though not the null result of the linear control.

4.6. Where it fails: fringe slant

Every result above is for an unslanted grating, and that restriction is not a convenience. With everything else fixed, the paired gain depends strongly and non-monotonically on the fringe slant

ϕ (Fig. 3(b); Supplement 1, Section S3). At $\phi = 0°$, 5°, 10° and 20° it is 0.913, 2.077, 0.089 and 0.024 dB at $K = 1.96$ rad/µm and 0.899, 2.045, 0.185 and 0.011 dB at $K = 3.93$ rad/µm. The gain rises to a peak at a small slant that differs between the two K, and from $\phi = 7.5°$ onwards it has collapsed at both, with no value above 0.29 dB at any larger angle.

Mechanism. Both features are Bragg matching, and neither is a property of the exposure design. The slab is read out at normal incidence, where an unslanted grating is off Bragg. Tilting the fringes by ϕ rotates the grating vector, and the tilt that brings a normally incident readout onto the Bragg condition follows from $2\mathbf{k}_{\text{inc}} \cdot \mathbf{K} + |\mathbf{K}|^2 = 0$ as

$$\sin \phi_\star = \frac{K\lambda}{4\pi n_0}, \tag{8}$$

which is the Bragg angle itself: tilting the fringes by θ_B does what tilting the beam by θ_B would. That predicts $\phi_\star = 2.42°$ at $K = 1.96$ and 4.84° at $K = 3.93$ rad/µm, with no fitted quantity. The measured gain peaks at 2.5° and 5.0° (Fig. 3(b)), the two tested angles nearest those predictions, and the first-order diffracted fraction peaks at the same two angles (Fig. 3(c)), rising from 0.24 to 0.48 at $K = 1.96$ and from 0.02 to 0.52 at $K = 3.93$ rad/µm. Past $\phi_\star$ the grating detunes again: by $\phi = 7.5°$ the first-order fraction is 0.017 and 0.006. A reconstruction carried almost entirely by the undiffracted beam leaves nothing for an exposure design to improve, so the paired difference between two designs goes to zero.

The readout is validated at slant, directly. The agreement above is not a scalar-model artifact. We compared the split-step beam propagation used inside the design loop against RCWA on the same gratings, over 84 cases spanning both tested K, three index-modulation levels, every slant in the sweep and both incidence conventions, matching the illumination and the order bookkeeping explicitly and normalizing both engines the same way, so that the Fresnel losses present in RCWA but absent from the beam propagation cannot enter the comparison (Supplement 1, Section S11). At normal incidence, the geometry the design loop reads out in, the two engines agree on the first-order fraction to a median of 0.001 and at worst 0.027, with the slanted cases no worse than the unslanted ones (0.027 against 0.014). At Bragg incidence the worst deviation over the whole grid is 0.037.

This is a sharper result than the closed-form check supports, and it separates two things the earlier Kogelnik comparison could not. The closed form carries no slant term, so it fails as soon as the fringes tilt: its median deviation from RCWA jumps from 0.05 at $\phi = 0°$ to 0.41 at $\phi = 2.5°$ over a 336-case sweep and stays there. The beam propagation does model the per-slice shear, and tracks RCWA across the same range. The restriction of the headline results to unslanted gratings therefore rests on the recording model and the benchmark, not on the readout, which is checked here across the full slant range.

4.7. Robustness to twin miscalibration

One parameter. Here both exposures are designed once with the nominal parameters and recorded, without re-optimization, on a twin with D_0, σ, κ or $\Delta n_{\max}$ perturbed by ±10, ±25 or ±50% (Supplement 1, Section S4 and Fig. S6). BSGD's design uses no recording parameter (only $\Delta n_{\max}$, as the scale of its linear model), so the test measures how much MIL's advantage depends on a correct twin. The K-averaged gain ranges from 1.17 to 1.92 dB (nominal 1.81 dB). We swept $\Delta n_{\max}$ further because two fits of the twin to Bayfol HX data disagree on it by 5.7× (Section 5): the gain is 0.38 dB at −83%, 1.66 dB at +50%, 0.99 dB at +100% and -3.73 dB at +473%. A twin that underestimates the medium's index budget several-fold can therefore make MIL worse than the media-blind baseline; where between +100% and +473% the sign changes was not resolved.

Joint and structural errors. In 30 Monte Carlo draws with all four parameters perturbed independently and uniformly within ±25% (same designed exposures), the K-averaged gain

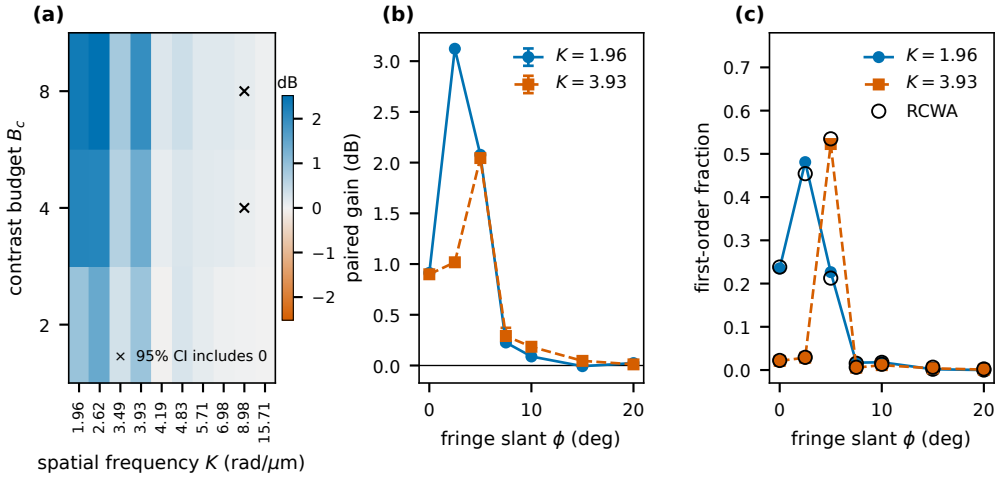


Fig. 3. Operating regime. (a) Paired gain over the 30 grid cells; crosses mark cells whose 95% interval includes zero. (b) Paired gain versus fringe slant at two K, $B_c = 2$, 95% t-intervals over three seeds. (c) First-order diffracted fraction over the same slant grid at normal incidence, the geometry the design loop reads out in, for $\Delta n = 3.5 \times 10^{-3}$: lines, the split-step beam propagation used in the design loop; open circles, RCWA. The gain in (b) peaks where the grating becomes Bragg-matched in (c), at the angle Eq. (8) predicts, and collapses once it detunes past it.

ranges from 1.61 to 1.94 dB (mean 1.76 dB, bootstrap 95% interval [1.73, 1.79]); 0 of 30 are negative. Beer–Lambert attenuation through depth (16 recorded slices) gives 1.81, 1.51 and 1.45 dB at optical density 0, 0.1 and 0.3. Re-optimizing both arms on media with D_0, σ or κ changed by up to ±50% ($K = 5.24$ rad/µm) gives 0.08 to 0.15 dB (nominal 0.11 dB).

Targets and noise. At $K = 3.93$ rad/µm and $B_c = 2$ the gain is 0.90 for bars, 3.22 for sparse random spots and 0.17 dB for a random binary target, and Gaussian detector noise at 5% of the peak changes the bar gain from 0.90 to 0.88 dB (Supplement 1, Section S8).

4.8. Which parameter has to be right

Robustness over a range is not the same as knowing which measurement to make carefully, so we rank the four recording parameters by how much the gain moves when each is wrong (Fig. 4). Two independent views agree on the order.

One at a time (Fig. 4(a)), the K-averaged gain spans 0.64 dB as $\Delta n_{\max}$ alone moves over ±50%, against 0.19 dB for κ, 0.19 dB for D_0 and 0.06 dB for σ, from a nominal 1.81 dB. The index budget therefore outranks the next parameter by 3.4×, and its worst case is an underestimate: at −50% the gain falls to 1.17 dB. For D_0 the worst case is an overestimate (50%); for κ and σ it is an underestimate.

Across the 30 joint draws (Fig. 4(b)), regressing the per-draw gain on the four standardized errors gives +0.071 dB per standard deviation for $\Delta n_{\max}$ [+0.054, +0.090], +0.024 for κ [+0.013, +0.034], -0.019 for D_0 [-0.031, -0.007] and +0.010 for σ [-0.003, +0.027]; 3 of the four intervals exclude zero, σ being the one that does not. The linear terms account for $R^2 = 0.88$ of the spread across draws. With this many draws the interaction and higher-order terms are not identifiable, so this is a first-order attribution over the tested ±25% box, not a Sobol decomposition; the rank order, not the coefficient values, is what we rely on.

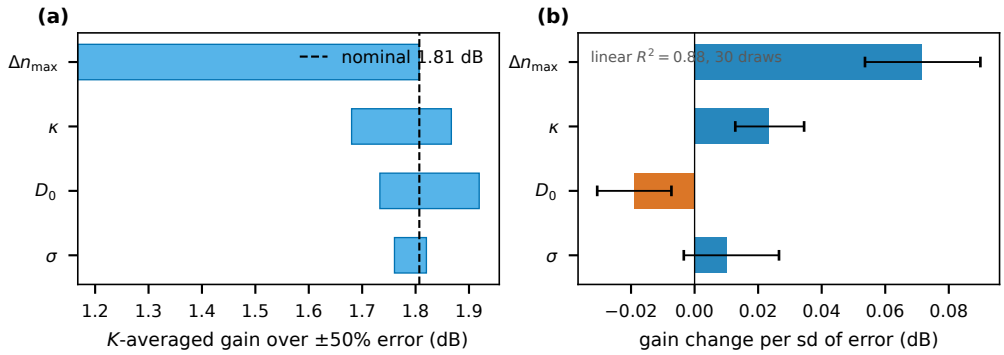


Fig. 4. Parameter sensitivity of the design gain. (a) Span of K-averaged paired gain covered as each parameter alone is varied over ±50%, against the nominal value (dashed). The worst case is the −50% end for Δn_{max}, κ and σ and the +50% end for D_0. (b) Gain change per standard deviation of each parameter's error across 30 joint draws, with bootstrap 95% intervals; colour marks sign. Rows are in the same order in both panels.

5. Twin validation against published data

We fit the twin to the first-harmonic index modulation Δn_1 versus dose in Fig. 3 of Bruder et al. [5] (Bayfol HX, $\Lambda = 700\,\text{nm}$, $K = 8.98\,\text{rad/µm}$, inside the tested grid): the source's kinetic-model curve ($16.7\,\text{mW/cm}^2$) and its measurement ($19.2\,\text{mW/cm}^2$), read visually rather than with a calibrated digitizer (about ±10% in dose, ±3–5% in Δn_1). Unfitted parameters take Bayfol HX values from the released configuration; each fit is the best of 8–10 random starts, and NRMSE is the root-mean-square residual over the range of the fitted data.

Fits and held-out prediction. With κ, Δn_{max} and k_{bleach} free, the twin reproduces the rise-then-decline shape of both series (Fig. 5; in-sample NRMSE 0.02 for the model curve, 0.17 for the measurement). Both are plotted against dose and the twin is dose-reciprocal at $\gamma = 1$, so a fit to one series predicts the other without refitting, with NRMSE 0.40 (model to measurement) and 0.31 (the reverse). The fitted k_{bleach} are 0.0667 and 0.0215 (default 0.2) and the fitted Δn_{max} 0.0418 and 0.0321. With k_{bleach} held at 0.2 the in-sample NRMSE is 0.17 and 0.28, but the measured-series optimum is degenerate (its κ saturates the medium before the first measured dose, giving an almost constant Δn_1; Supplement 1, Fig. S5), the held-out NRMSEs are 0.48 and 1.09, and the fitted Δn_{max} disagree by 5.7×, which sets the range of the Δn_{max} sweep in Section 4.7. Fig. S5 also shows a fit to a PQ/PMMA $\Delta n_1(t)$ model curve [6] outside the tested K range (NRMSE 0.10).

What this establishes, and what it does not. The model family can reproduce, and predict across two views of the same condition, the dose response of one commercial photopolymer with effective parameters. It does not pin the default medium of Table 1 to a real material or test transfer across K or media: the two series share a material, a grating and nearly the same intensity, and one is a model output. D_0 and σ are identifiable only from multi-K growth-curve families, which we did not have; Section 6 states what to measure instead. The headline claims therefore rest on the paired design and the miscalibration tests, not on absolute agreement with a real medium.

6. Design guidelines and calibration protocol

This section is the operational summary: what a laboratory would measure, in what order, and when not to bother.

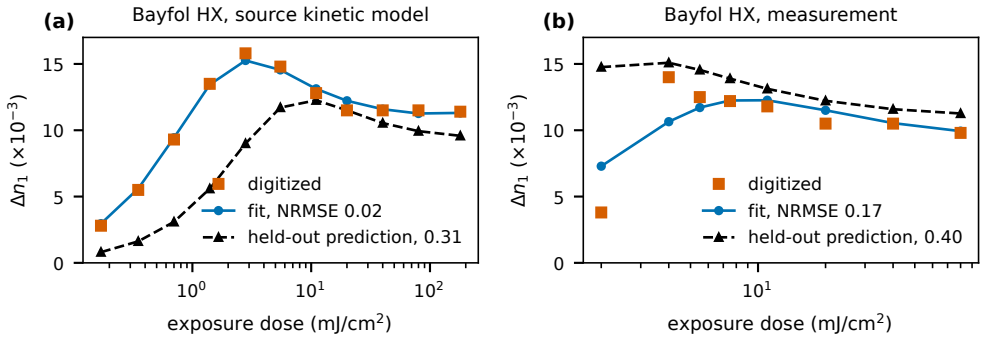


Fig. 5. Twin fitted to Δn_1 digitized from Bruder et al. [5] (Bayfol HX): (a) the source's kinetic-model curve, (b) its measurement. Solid, fit to the series shown (κ, $\Delta n_{\max}$, k_{bleach} free); dashed, parameter-free prediction from the fit to the other series. Legends give NRMSE.

Step 1: decide whether the geometry qualifies. The results support unslanted transmission gratings, which is where the recording model and the benchmark were run. Fringe slant is not a free parameter: the gain depends on it through Bragg matching, peaking near $\phi_\star = \arcsin(K\lambda/4\pi n_0)$ and collapsing from $\phi = 7.5°$ (Section 4.6). A setup that cannot hold the slant to well inside that window should not expect the gains reported here. Within the qualifying geometry, read Fig. 3(a): the benefit is concentrated below about 3 rad/µm and grows with the contrast budget the setup can deliver. At the top of the tested range the gain is under 0.05 dB and not worth the calibration effort.

Step 2: measure the dose response at the working point. One Δn_1 versus dose curve, recorded at the spatial frequency and intensity the application will use, identifies κ and $\Delta n_{\max}$. Fit k_{bleach} jointly rather than assuming it: holding it at a nominal value made our own fit degenerate, with κ saturating the medium before the first measured dose (Supplement 1, Section S9). Take $\Delta n_{\max}$ from the same experimental condition as the curve being fitted, never from a separate experiment on nominally the same material; two fits that differ in this respect disagreed by 5.7× (Section 5).

Step 3: measure at more than one spatial frequency. D_0 and σ act only through the spatial-frequency dependence, so a single-K curve cannot identify them and should not be used to try. A growth-curve family at three or more spatial frequencies spanning the intended operating range, or an independent measurement of monomer diffusivity, is the minimum. Thickness and shrinkage are measured separately by angular scan.

Step 4: spend the effort where it matters. The ranking of Section 4.8 is unambiguous: $\Delta n_{\max}$ dominates, outranking κ by 3.4× and σ by more than an order of magnitude. A 50% underestimate of the index budget costs 0.64 dB of the 1.81 dB nominal gain, and a several-fold underestimate reverses the sign of the benefit entirely (Section 4.7). An error in σ, by contrast, is the one whose effect is not resolved from zero across the joint draws. If calibration time is limited, spend it on the index budget and the dose response, not on the non-local length.

Step 5: budget the compute. Designing through the model cost about 21.7× per iteration here, and the advantage survives giving the media-blind baseline that factor more iterations (Section 4.4). At the grid size used, a design is minutes on a laptop GPU, which is small against the cost of a wasted film.

When not to use the method. If the recording geometry is slanted, since nothing here was run at $\phi \neq 0$ beyond the sweep itself; if the target spatial frequency is at the top of the tested range; if $\Delta n_{\max}$ is not known to better than a factor of two under the recording condition; or if the

application sits at the very lowest spatial frequencies and a calibrated pointwise saturating model is already available, since that surrogate recovers 36–44% of the gain there (Section 4.3) for a fraction of the modelling effort.

7. Limitations

No physical recording. All results are simulation outputs of one model family. Whether the gains transfer to a physical photopolymer is untested; the decisive experiment, recording optimized and media-blind exposures in a characterized medium and comparing the measured reconstructions, is future work. What the multi-parameter miscalibration and sensitivity studies do cover is the question a laboratory would ask next: how wrong the model can be before the benefit disappears, and which parameter that depends on.

Validation breadth. The twin is compared with one commercial medium at one spatial frequency, in two views of the same recording condition, one of which is itself a model output, and the points were read visually rather than with a calibrated digitizer. Transfer across K and across media is untested, which is exactly the measurement Step 3 of Section 6 asks a calibrating laboratory to make.

Geometry and targets. One-dimensional transverse recording of binary bar targets in unslanted gratings, with one default medium and three seeds per cell. The readout itself is not the limitation here: the beam propagation used in the design loop agrees with RCWA [20] (computed with TORCWA [27]) to at worst 0.037 in first-order fraction across the slant range (Section 4.6). The closed-form Kogelnik reference is the weaker tier: over a 90-case grid of K, Δn, polarization and slant it deviates by at most 0.29 in diffraction efficiency for unslanted geometries and by up to 0.98 at a 20° slant (Supplement 1, Section S5), and it is used only as a cross-check, never inside the design loop. What restricts the headline results to unslanted gratings is therefore the recording model and the benchmark, which were run at $\phi = 0$ throughout, not the propagator.

Model and statistics. The recorded profile is uniform through depth except in the Beer–Lambert check; diffusion uses a spatially averaged coefficient; the index map and its shape constant are modelling choices; shrinkage is a placeholder; the medium parameters are illustrative; and the factorial covers interactions among the saturating mechanisms only. Three seeds per cell measure initialization sensitivity, which is small here; model and parameter error are probed only by the miscalibration and sensitivity studies over the stated ranges. The 10-point grid does not resolve features between grid points, such as the zero crossing at $B_c = 2$.

8. Conclusion

In simulation, designing the exposure of a write-once volume photopolymer through a differentiable NPDD and beam-propagation model improves on media-blind design in 29 of 30 unslanted grating cells, by up to 2.52 dB at low spatial frequency, and removes the exposures that fall more than 2 dB short of an oracle reference. The gain requires a nonlinear dose response but no particular saturating mechanism, which means a laboratory need not identify its medium's nonlinearity, only model it. It decays toward zero at high spatial frequency, survives joint parameter errors of ±25%, and depends above all on the recorded index budget $\Delta n_{\max}$, a several-fold underestimate of which reverses the benefit. Its dependence on fringe slant is optical rather than algorithmic: the gain peaks where a tilt of $\arcsin(K\lambda/4\pi n_0)$ brings the normally illuminated grating onto Bragg, and collapses once it detunes past that angle, a correspondence confirmed against RCWA. The calibration protocol of Section 6 states the minimum measurement set; testing these predictions in a recorded medium is the next step.

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points digitized in Section 5, and to draft and edit the text; the author reviewed all outputs and takes full responsibility for the content.

Disclosures. The author declares no conflicts of interest.

Data availability. Code, experiment manifests, raw per-job result files and the scripts that generate every number and figure in this paper are available on GitHub (https://github.com/ultramagnus23/HoloForge) and archived in Ref. [28].

Supplemental document. See Supplement 1 for supporting content.

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